Mark **Edmonds**

Staff AI/ML Applied Scientist · Foundation Models for Autonomous Driving · Imitation & Reinforcement Learning

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Summary

Staff AI/ML applied scientist building foundation models for autonomous driving. At General Motors, I develop end-to-end driving models trained with imitation and self-play reinforcement learning. At Cruise, I co-developed DriveGPT (ICML 2025), which demonstrated data and model scaling laws for autoregressive driving behavior models, and distilled large models for on-vehicle deployment. PhD in Computer Science from UCLA on causal learning, imitation learning, and explainable AI, with work published in Science Robotics, ICML, CVPR, and AAAI.

Experience

Staff AI/ML Applied Scientist, General Motors, Remote — Feb 2026 - Present

- Develop end-to-end driving models trained with imitation and self-play reinforcement learning.
- Joined GM through its acquisition of Cruise.

Senior Applied Scientist II, Cruise, Remote — Apr 2024 - Feb 2026

- Co-developed DriveGPT, scaling autoregressive behavior models for driving to 1.4B parameters and 120M training segments (∼50x more data than prior published models); demonstrated data and model scaling laws for driving (ICML 2025).
- Applied imitation and reinforcement learning to trajectory generation for trajectory planning.
- Distilled large behavior models into compact models for on-vehicle deployment.
- Co-inventor on a patent application for autoregressive models for autonomous agents (US 2026/0131822 A1).

Senior Applied Scientist, Cruise, Remote — Oct 2021 - Apr 2024

- Worked at the intersection of machine learning and motion planning on the Behavior Planning team, starting with maneuver planning.
- Applied imitation learning to trajectory generation for behavior planning; used PyTorch and ROS to deploy safer, more capable ML models to vehicles driving on public roads.

Graduate Student Researcher, Computer Science Department, UCLA, Los Angeles, CA — Sep 2016 - Sep 2021

- **Causal learning**: built OpenLock, an open-source virtual escape room in which every room shares an abstract causal structure (a sequence of levers) while surface features change, to test how humans and AI transfer causal knowledge. Ran human-subject experiments showing people discover and transfer the abstract structure, and developed a hierarchical Bayesian model combining abstract-level structure learning with instance-level induction that matched human transfer, while seven model-free RL algorithms failed (AAAI 2020 oral; CogSci 2018 oral; CogSci 2019).
- **Imitation learning**: taught a robot to open medicine bottles from human demonstrations by integrating force and pose; captured hand forces and poses with a custom tactile glove (IMUs and force sensors) and planned robot actions with an embodied haptic model and a stochastic grammar (IROS 2017, two papers).
- **Explainable AI**: extended the bottle-opening robot to explain its own behavior; showed that the explanations that best foster human trust differ from the components that drive task performance (Science Robotics 2019; covered by IEEE Spectrum).
- Also contributed to abstract causal reasoning (CVPR 2021) and in situ human-robot value alignment (Science Robotics 2022).

Undergraduate Researcher, Air Force Research Lab (AFRL), Dayton, OH — May 2014 - Sep 2015

- Conducted Declarative Memory (semantic knowledge retrieval system) research for the ACT-R cognitive architecture.
- Architected a new declarative memory system using CUDA, thread pools, parsers, and inter-process communication (IPC).
- Continued project work between summers as undergrad thesis research.
- Result: Parallelized declarative retrievals; yielded a 100x speedup over the fastest existing implementation.

Selected Publications

- X. Huang, E.M. Wolff, P. Vernaza, T. Phan-Minh, H. Chen, D.S. Hayden, **M. Edmonds**, et al. "DriveGPT: Scaling Autoregressive Behavior Models for Driving." *ICML 2025*.
- L. Yuan*, X. Gao*, Z. Zheng*, **M. Edmonds**, et al. "In situ bidirectional human-robot value alignment." *Science Robotics 2022*.
- C. Zhang, B. Jia, **M. Edmonds**, S.C. Zhu, Y. Zhu. "ACRE: Abstract Causal REasoning Beyond Covariation." *CVPR 2021*.
- **M. Edmonds**, X. Ma, S. Qi, Y. Zhu, H. Lu, S.C. Zhu. "Theory-based Causal Transfer: Integrating Instance-level Induction and Abstract-level Structure Learning." *AAAI 2020* (oral).
- **M. Edmonds***, F. Gao*, H. Liu*, X. Xie*, et al. "A tale of two explanations: Enhancing human trust by explaining robot behavior." *Science Robotics 2019*. Covered by IEEE Spectrum and The Conversation.

- Patent application: "Autoregressive Models for Autonomous Agents." US 2026/0131822 A1 (2026).
- 4 journal and 10 conference papers in total (* equal contribution). Full list: mjedmonds.com or Google Scholar.

Education

University of California, Los Angeles (UCLA), Los Angeles, CA
Ph.D. in Computer Science, Artificial Intelligence Concentration; Advisor: Prof. Song-Chun Zhu — Sep 2021
M.S. in Computer Science — Jun 2017

University of Dayton, Dayton, OH
B.S. in Computer Engineering; Magna Cum Laude — May 2015

Skills

Tools Python, C++, PyTorch, TensorFlow, CUDA, ROS
Methods Foundation models, Transformers, Scaling laws, Imitation learning, Reinforcement learning (incl. self-play), Model distillation, Behavior prediction and motion planning, Autonomous driving, Robotics, Causal learning, Explainable AI, Bayesian models

Teaching Experience

Adjunct Professor, Computer Science Department, Santa Monica College, Santa Monica, CA — Jun 2016 - Sep 2021

- Instructed 25 courses: Internet Programming (HTML, CSS, JavaScript, MySQL, and PHP), Intro to C, and Intro to C++.

Teaching Assistant, Computer Science Department, UCLA, Los Angeles, CA — Sep 2015 - Jun 2016

- Teaching assistant for Introduction to C, Introduction to C++: led discussion hours (~50 students), held office hours.

Honors, Talks & Press

Keynote speaker, RSS 2018 Causal Imitation Workshop, Pittsburgh, PA — 2018
Oral presentations, AAAI 2020 and CogSci 2018, New York and Madison — 2018--2020
Press for Science Robotics paper, IEEE Spectrum, The Conversation, Tech Xplore, Online — 2019--2020
Invited speaker, ONR MURI meetings, New Hampshire and Los Angeles — 2017--2018
NSF Doctoral Consortium, IROS 2017, Vancouver, BC — 2017
Anthony Horvath and Elmer Steger Award of Excellence, University of Dayton, Dayton, OH — 2015